

Wide background with 6+ years expertise in Python, R, Java, SQL, statistical Analysis, deep understanding of data structures and algorithms, and applying that background toward data collection/cleaning/analysis, business analysis, and improve competitiveness:

- Deep experience in ML, DL and Cloud technologies (TensorFlow, PyTorch, Google Jax) for dev & deployment.
- Experienced in SQL, traditional programming languages including Python, R, LLMs with a focus on scalable solutions for Big Data using Apache Spark Machine Learning (Databricks) and building data models for forecasting.
- Designed & deployed local AI agents using the Vicuna language for secure & cost-efficient tasks, e.g., SQL query generation, optimization, forecasting using both ML/DL techniques to ensure data privacy and enhancing prediction.
- Built a Python pipeline for Retrieval-Augmented Generation (RAG) using Langchain to generate context-aware marketing content. Additionally, leveraged a Frontend Framework (Streamlit for visualization) in a Web application development, and leveraged OpenAI's GPT-3 model at the Backend for AI-powered marketing assistance.
- Built a customer sentiment analysis (NLP) about airlines. Skilled in AWS data analytics including RDS and DynamicDB, utilizing Redshift, Athena, and QuickSight for scalable pipelines and interactive dashboards.
- Utilized OpenCV to implement image classification and facial masking techniques to anonymize individuals in visual data, safeguarding privacy while enabling data analysis.
- In multiple applications, I built-in monitoring tools to collect and generate alerts when appropriate.

Experienced in all stages of development lifecycle, from data preparation, monitoring, to deployment, including model inference using BigQuery and PyTorch.

SKILLS

Programming Languages: Python, SQL, R, Perl, JavaScript, MATLAB, Bash

Data Analysis Tools: Microsoft Excel, Tableau, Streamlit, Jupyter Notebook, MATLAB IDE

Cloud Technologies: Amazon QuickSight, Amazon Redshift, Amazon Glue, AWS, Azure, Databricks, Spark

Machine Learning and Deep Learning Frameworks: TensorFlow, PyTorch, Google Jax, Pandas, Scikit Learn,

Statistics libraries: Statsmodel, NumPy, SciPy, etc.

Others: Git, Docker, GPT, Problem solving, and economics

WORK EXPERIENCE

BIG DATA INSIGHTS • Cupertino, CA • 7/2023- Present

DATA SCIENTIST

- Researched industry trends and competitor practices in sales forecasting to ensure an optimal solution for clients.
- Curated historical sales data and applied advanced feature engineering to enhance model performance & accuracy.
- Used statistical models to capture long-term trends and seasonality, while employing k-NN to adjust predictions based on historical analogs and nonlinear patterns.
- Leveraged machine learning expertise with algorithms like ARIMA, SARIMA, and LSTMs to address sales forecasting challenges, and developing forecasting models and applications.
- Implemented N-BEATS neural networks for accurate and robust sales forecasting, significantly improving model performance and prediction accuracy.
- Implemented a CI/CD pipeline with Jenkins for time series, data preprocessing, model training, and accuracy validation.
- Utilized techniques such as PELT and Binary Segmentation algorithms for change point detection to identify significant deviations between predicted and actual sales, enabling uncovering potential structural shifts in sales behavior.
- Employed causal inference techniques such as propensity score matching and double ML to assess the impact of external factors (promotions, competitor pricing, seasonality) on sales deviations, and enhanced forecasting accuracy.

SANGU • Sunnyvale, CA • 7/2023- 10/2023 (Part-Time)

DATA ANALYST

- Orchestrated seamless data importing into a PostgreSQL database, enabling advanced analysis, and significantly contributing to informed decision-making and strategic planning initiatives.
- Leveraged remote access to Linux servers to extract valuable data from log files, employing Perl for efficient data scraping and consolidation, thereby optimizing data retrieval processes.
- Played a pivotal role in migrating crucial customer data into a CRM system, facilitating enhanced relationship management and bolstering customer service endeavors.

AMAZON, LAB 126 • Sunnyvale, CA • 9/2021- 7/2023

DATA COLLECTION SPECIALIST, LAB ENGINEER

- Developed **Python** scripts to elevate data analysis and problem-solving capabilities, encompassing script execution, debugging, and optimization. Conducted meticulous data collection and cleaning from diverse audio devices to ensure exceptional **data quality** and **reliability**.
- Designed and implemented effective data validation strategies and frameworks to identify and address data quality issues, such as **outliers**, **missing values**, and inconsistencies.
- Developed a **BASH script** to systematically validate file naming conventions across multiple folders, ensuring adherence to the prescribed standards and streamlining file organization and management.

BIG DATA INSIGHTS • Cupertino, CA • 8/2019 - 8/21

DATA SCIENTIST

- Developed a **K-means clustering** algorithm to identify optimal event hosting location based on the population demographics. This helps facilitating data-driven decision-making for successful event planning and execution.
- Researched improving our product competitiveness in a distributed system, with focus on minimizing response-time. Worked closely with the development team to incorporate my findings.
- Conducted competitive analysis to ensure the sustained competitiveness of the product, analyzing market trends, customer needs, and industry benchmarks to inform strategic decision-making and drive product enhancements.

LCI/VERISK • Burlingame, CA • 12/2018 – 8/2019

Operations & Data Analyst

- Developing and implementing **SQL** programs to efficiently manage and modify **database** tables, ensuring optimized **data processing** and seamless integration within the data science workflow.
- Effectively employed proprietary software to perform comprehensive audits of client records, ensuring meticulous examination of data integrity and facilitating actionable insights
- Collaborated with cross-functional teams, leveraging advanced **MS Excel** skills and pivot tables, to analyze and manage data effectively for **reporting** purposes, contributing to the overall success of projects.

EDUCATION

- **Data Science Certificate (1.5 years) with Specialization in Machine and Deep Learning**, Springboard:6/2020-11/2021. Courses covered include: Python, Data Structures and Algorithms, OS, RDBMS, Machine Learning, Deep Learning, Image Classification and Processing, AI Tools and Libraries, hands-on application projects development.
- **M.B.A. in Business Analytics and Technical Marketing**, St. Mary's College of California: 2015-2018
- **Bachelor of Science in Economics**, Santa Clara University: 2008-2013. Worked as Information Technology Assistant (1/2011 – 8/2012)

ACADEMIC PROJECTS

Below are some of the projects posted at GitHub (<https://github.com/AdamEzzat1>):

1. Finding Best Location for Sporting events, using Clustering, based on demographics. ([GitHub Link](#))

- Techniques: K-means clustering
- Tech Stack: Python, Google Colab, pandas, numpy, matplotlib, sklearn, and geopy
- Key Accomplishment: used k-means clustering & geography to find the best location for sport events.

2. Analyzing Twitter posts for Sentiment analysis and Topic modeling ([GitHub Link](#))

- Techniques: Logistic Regression, Random Forest Classifier
- Tech Stack: Python, Google Colab, Pandas, Numpy, NLTK, Textstat, Matplotlib, Altair, Sklearn, and Keras
- Key Accomplishment: Leveraged Kaggle datasets to scrutinize Twitter posts pertaining to diverse airlines. Employed the tweets to conduct a comprehensive sentiment analysis, coupled with determining the most influential terms contributing to positive and negative sentiment.

